Internet of Things (IOT)

A type of network system in which physical devices such as sensors, gadgets, vehicles, and home appliances are connected to the internet, allowing them to exchange data.

* Functional Requirement:

(i) Scalability:
The system should remain efficient even as the number of devices increases.

(ii) Interoperability:- Device from different manufacturer should be able to communicate with each other.

(iii) Security:- Data should be secure, and unauthorized access should not be allowed.

(iv) Real-time monitoring:- The system should allow real-time monitoring and analysis of data.

(v) Low-Power Consumption:- Devices should use minimal power.

* Architecture of IOT:- (4 layers)

(i) Perception layer:- Collect data (eg:- Sensors, RFID tags, Cameras)

(ii) Network layer:- Transmit data (eg:- Internet, wifi, Bluetooth)

(iii) Middle ware layer:- Handles decision making.
(eg:- Data processing, filtering data)

(iv) Application layer:- Provides user-specific services.
(eg:- Smart phones)

* Web 3.0 :-

→ It is a Semantic web in which machines are developed
→ to understand and interpret data like human beings.
→ Web 3.0 + IoT enables smart decision making machines without human intervention.
→ Uses include blockchain-enabled IoT Security and dAPPs (decentralised Application).

* VIEWS of IoT :-

(a) Technology oriented view :-
→ Sensors, Actuators, cloud computing, Big Data & AI.

(b) Application oriented view :-
→ Smart phones, Healthcare, Agriculture.

(c) Service oriented view :-
→ IOTaaS (as a Service)

* Ubiquitos of IoT :- (Present everywhere)

→ Smart ~~phone~~homes = To control automatically (Light AC, Machine etc)
→ Smart cities = Traffic management, Pollution Control
→ Health care = fitness Bands, Remote patient monitoring.
→ Agriculture = Soil monitoring, Smart irrigation.
→ Industrial IoT = machines, Real time monitoring. (IIoT)
→ Retail = Inventory tracking, Employee Biometric, Smart checkout.

* Four pillars of IoT:-

(i) People (ii) Process (iii) Data (iv) Things.

* DNA of IoT:-

Data, Network, Application

* Middleware in IoT:-

The software layer that manages Data between Sensors and Application.

Functions: (A) Device management.
(B) Data management.
(C) Security Handling.
(D) Interoperability

* Protocol used:-

(i) MQTT (Message Queuing Telemetry Transport)
↳ light weight low Bandwidth

(ii) CoAP (Constrained Application Protocol)
↳ Optimised web transfer protocol.

(iii) HTTP/HTTPS
↳ Traditional Internet communication.

(iv) AMQP:- (Advanced messaging Queuing Protocol)

* Protocol Standardization:-

→ To develop a common set of ~~rules~~ rules, which all IoT devices can follow.

→ This is necessary so that devices from different manufacturers can communicate with each other.

* Objective:-

→ To ensure interoperability b/w different devices.

→ To bring ~~and~~ Scalability and flexibility.

→ To maintain Security and data integrity.

* Efforts for Standardization:-

(i) IEEE:-(Institute of Electrical & Electronics Engineers)

→ Defined protocols for the Physical & Data link layers.

(ii) IETF:-(Internet Engineering Task Force)

→ Develop Network layer protocols, such as 6LoWPAN (low Powered wireless Personal Area Network)

(iii) ETSI:-(European Telecommunication Standard Institute)

→ Standardizes M2M (machine to machine) protocols.

* machine to machine protocols:-

→ machine communicate directly or through a gateway.

eg MQTT (Message Queue Telemetry Transport)

CoAP (Constrained Application protocol)

LwM2M (Lightweight machine to machine

* WSN Protocols :-

→ Wireless Sensor Network.
→ Sensors Collect and transmit data.
Ex: Zigbee, 6LoWPAN, RPL.

* IEEE 802.15.4 :-

→ Zigbee
→ Basic protocol for low Rate wireless Personal Area Network.
→ works on 2.4 GHz / 868 MHz / 915 MHz.
→ Short range & reliable transmission.

* WoT (web of things)

→ Extension of IoT.
→ It uses web technologies (HTTP, JSon) to allow devices to be easily accessed and integrated.
→ It is the abstraction layer build on top of IoT, So that all devices can communicate using web standards and protocols.
Ex: If a smart bulb has a URL, It can be turned on/off directly from a web browser.

* ~~Features~~ :-

* SCADA & RFID:-

→ SCADA (Supervisory Control and Data Acquisition)
→ used for Industrial Control System.
→ Used in electricity grids & Water Systems.
→ DNP (Distributed Network protocol) is used in SCADA.
→ RFID (Radio frequency Identification)
→ Used for wireless identification.
→ Used for Communication between Tags (like fastag) & readers.

* IoT & WoT.

Aspects	IoT	WoT
Connectivity	Device Centric & physical connectivity	Web-centric & application layer abstraction.
Protocol	MQTT, CoAP, Zigbee, bluetooth	HTTP & JSON
Data formats	Binary & lightweight	JSON & XML supported
Interface	Mostly Proprietary or protocol specific	Standard web Interface (URI)
Interoperability	low due to diverse protocol	high via web Standard
Goal.	To connect real world objects	To embed IoT devices into web

* CoT (cloud of things)
→ cloud computing + Internet of things.
→ Data from IoT Devices is accessed, stored, processed and managed through platforms.

* Grid Computing:
→ A distributed computing model in which different resources (servers & storage) are connected via a network to perform large-scale computation.
→ Resources are shared in this model.

* SoA (Service Oriental Architecture)
→ An architectural pattern that defines services.
→ Different services are independent and communicate with each other via messages.
→ In SoA, cloud platforms are used to increase scalability and flexibility.

* Cloud Computing:-
→ A model in which computing resources.
→ cloud standard are important in cloud computing because they ensure interoperability, security and portability.

Ex
- OCCI (open cloud computing Interface)
- CSA (cloud Security Alliance)
- ISO (International Standard organization)

* MCC (Mobile cloud computing)

→ Connecting mobiles through Cloud platforms.
  Eg: icloud, Google Drive, Dropbox.

→ Folder Sync.

* IIoT :-

→ Industrial Internet of things.

→ In industrial environments, sensors, devices, and machines are connected to the Internet for real-time monitoring.

→ These devices collect and share data in real-time so that industries can optimize their operations.

→ IIoT 4.0 is the fourth Industrial Revolution, in which physical machines are integrated with digital technologies.

→ Its main focus is on automation, data exchange, and integrating other technologies.

* Basic Technologies used in IoT :-

1) Sensors & Actuators (collect real time data)
2) cloud computing
3) Big data
4) AI & machine learning
5) 5G Network.

* ARMET :-

→ Autonomous Risk management & Engineering Technology
→ Used to make IIoT systems secure and risk-free.
→ This approach integrates Safety and Security. In which AI & machine learning are used.

* Role of IoS in collaborative production Environment:

(i) Increased Autonomy and Agility.
(ii) Resource management.
(iii) Synchronization
(iv) Software agents